

# Modelling Volatility III: Variants of the GARCH Model

Week 2, Day 3 – Advanced Financial Econometrics

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References: Tsay, *Analysis of Financial Time Series*, Ch. 3 – IGARCH (Engle–Bollerslev, 1986), GARCH-M (Engle–Lilien–Robins, 1987), EGARCH (Nelson, 1991).

All empirical results use the course `data/` folder: S&P 500 daily returns, 2000–2026.

## Where we were, and where we go today

- **So far.** ARCH made variance move; **GARCH(1,1)** gave it long memory cheaply:  
 $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$ .
- GARCH(1,1) is the workhorse – but it makes *three* strong assumptions that the data quietly push back on.

### Today: three questions plain GARCH cannot answer

- ① Its persistence  $\hat{\alpha} + \hat{\beta}$  sits at 0.98–0.99. What if it is *exactly* 1? → **IGARCH**
- ② It assumes expected return is *unrelated* to risk. But shouldn't riskier days pay more? → **GARCH-M**
- ③ It treats a −3% and a +3% day *identically*. But crashes scare markets more than rallies. → **EGARCH**

*Three targeted fixes, one comparison table, all on real S&P 500 data.*

# Roadmap

- ➊ Three limitations of plain GARCH
- ➋ IGARCH: when shocks never fade
- ➌ GARCH-M: paying for risk
- ➍ EGARCH: bad news bites harder
- ➎ Putting it together: a comparison
- ➏ Beyond fixed parameters: regime switching
- ➐ Chapter 4 preview: nonlinear time series
- ➑ Practice

## Three limitations of plain GARCH

# Plain GARCH(1, 1), and its three blind spots

## The baseline (fitted daily S&P 500)

$$r_t = \mu + a_t, \quad a_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \underbrace{\omega}_{(0.025)} + \underbrace{\alpha}_{(0.118)} a_{t-1}^2 + \underbrace{\beta}_{(0.863)} \sigma_{t-1}^2.$$

## What it silently assumes – and today's fix for each

| Hidden assumption in GARCH              | The data say                                             | Fix            |
|-----------------------------------------|----------------------------------------------------------|----------------|
| $\alpha + \beta < 1$ (variance reverts) | $\hat{\alpha} + \hat{\beta} = 0.981$ , glacial reversion | <b>IGARCH</b>  |
| mean $\mu$ is a constant, risk-blind    | risk should be <i>paid</i> for                           | <b>GARCH-M</b> |
| $+a$ and $-a$ hit variance equally      | down days scare more                                     | <b>EGARCH</b>  |

Each variant changes *one* thing. Keep the fitted GARCH numbers above in mind as the yardstick.

IGARCH: when shocks never fade

# IGARCH – integrated GARCH

The one change: pin persistence at exactly 1

Impose  $\alpha + \beta = 1$  in the GARCH recursion. Writing  $\beta = \lambda$ ,  $\alpha = 1 - \lambda$  and (usually)  $\omega = 0$ :

$$\sigma_t^2 = (1 - \lambda) a_{t-1}^2 + \lambda \sigma_{t-1}^2.$$

- This is exactly the **exponentially weighted moving average** (EWMA) of squared returns – **RiskMetrics** uses  $\lambda = 0.94$  for daily data.
- **How it differs from GARCH:** with  $\alpha + \beta = 1$  the unconditional variance  $\omega/(1 - \alpha - \beta)$  is *infinite* – there is **no long-run level to revert to**. A shock to variance is **permanent**.
- **Fewer parameters:** one number  $\lambda$  (or none, if you fix 0.94). Simple and robust.

# IGARCH – which problem it addresses, and the catch

## Why anyone would want a non-stationary model

Estimated equity persistence is *so* close to 1 ( $\hat{\alpha} + \hat{\beta} = 0.981$  for the S&P, 0.992 for Apple) that mean reversion is nearly invisible over practical horizons. IGARCH stops pretending there is a stable long-run variance and just says: “**tomorrow’s best volatility guess is essentially today’s.**”

## Fitted to daily S&P 500

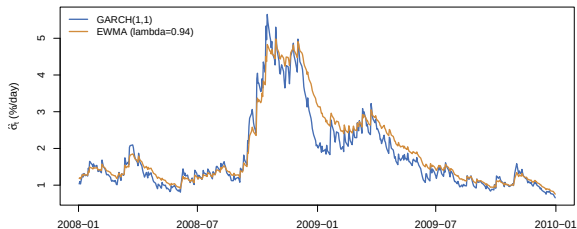
MLE gives  $\hat{\lambda} = \mathbf{0.93}$  (so  $\alpha = 0.07$ ,  $\beta = 0.93$ ), almost the RiskMetrics 0.94. Its in-sample volatility path is **96% correlated** with the GARCH path – *in sample they look the same.*

- **The catch is in forecasting:** because there is no anchor, the multi-step forecast does *not* revert – with  $\omega = 0$  it is dead **flat** at all horizons.
- So long-horizon VaR from IGARCH never “calms down.” Convenient and conservative, but it discards the mean-reversion GARCH worked to capture.

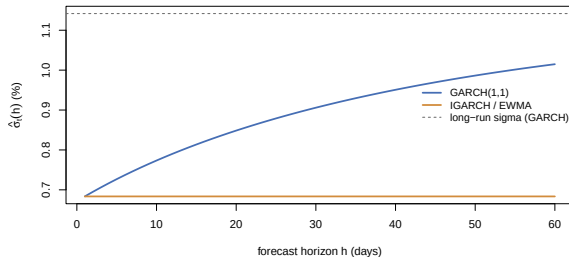


# IGARCH vs GARCH: identical in sample, different in forecast

In sample, IGARCH/EWMA and GARCH track closely; they differ only in FORECASTS



GARCH forecast mean-reverts; IGARCH (EWMA) never does



**Left:** through 2008–09 the EWMA/IGARCH ( $\lambda = 0.94$ ) and GARCH bands are almost on top of each other.  
**Right:** from today's origin, GARCH's forecast *reverts* to its long-run 1.14% while IGARCH stays **flat** – today's level, forever. *The whole difference lives in the forecast.*

GARCH-M: paying for risk

## GARCH-M – GARCH in mean

The one change: let volatility enter the *mean* equation

$$\begin{aligned}r_t &= \mu + \delta \sigma_t + a_t, & a_t &= \sigma_t \varepsilon_t, \\ \sigma_t^2 &= \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2.\end{aligned}$$

- Plain GARCH models the mean and the variance *separately*:  $\mu$  is a constant, indifferent to how turbulent the market is.
- **How it differs:** GARCH-M feeds the conditional volatility  $\sigma_t$  *back into* the expected return. Now  $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mu + \delta \sigma_t$  **rises on risky days**.
- **$\delta$  is the “price of risk”** – the extra expected return demanded per unit of volatility. (Some texts use  $\sigma_t^2$  or  $\log \sigma_t^2$  in place of  $\sigma_t$ .)

# GARCH-M – which problem it addresses

## The risk–return trade-off

Finance theory says investors must be *compensated* for bearing risk: periods of high volatility should carry higher *expected* returns. Plain GARCH cannot express this – its mean is flat. GARCH-M builds the trade-off directly into the model and **lets the data estimate its size,  $\delta$** .

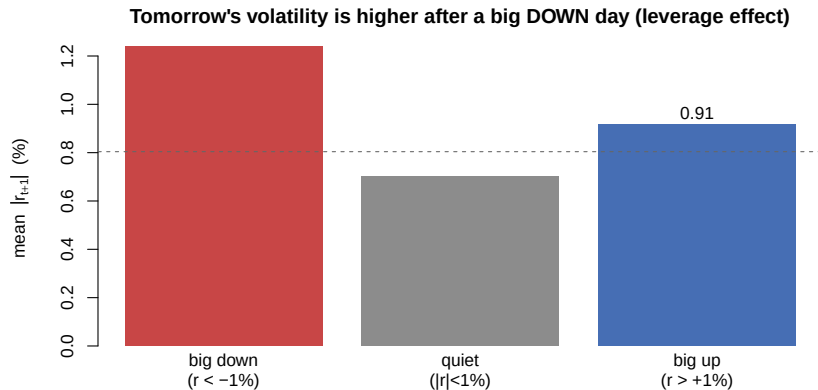
## Fitted to daily S&P 500

$$\hat{\delta} = \mathbf{0.108} \quad (\text{SE } 0.036, \ t = \mathbf{3.0}), \quad \hat{\alpha} = 0.121, \ \hat{\beta} = 0.860.$$

- $\hat{\delta} > 0$  **and significant**: a genuine, positive risk premium. When conditional volatility is one point higher (say  $\sigma_t$  from 1% to 2%/day), expected daily return rises by about 0.11% – the market pays more when it is scarier.
- The variance dynamics  $(\alpha, \beta)$  barely move: GARCH-M *adds* a mean channel, it does not disturb the volatility one.
- *Honest caveat*: this premium is notoriously fragile – weaker or insignificant in many samples and at the daily frequency. Here it is clean.

EGARCH: bad news bites harder

# The motivating fact: volatility is asymmetric



Daily S&P 500: the average size of *tomorrow's* return, split by what *today* did. After a **big down day** ( $r_t < -1\%$ ) tomorrow averages **1.24%**; after a **big up day** ( $r_t > +1\%$ ) only **0.92%**. Bad news raises future volatility *more* than equally-sized good news – the **leverage effect**. *Plain GARCH, which squares the shock, is blind to this: it sees only magnitude, never sign.*

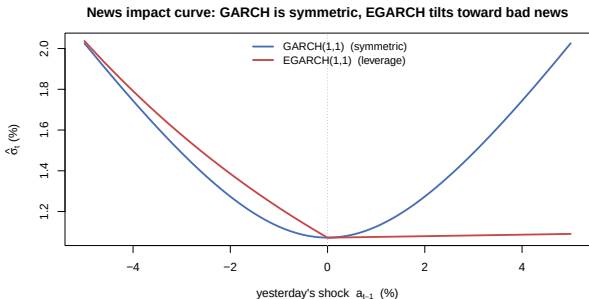
# EGARCH – exponential GARCH (Nelson, 1991)

Two changes: model  $\log \sigma_t^2$ , and add a *sign* term

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha(|z_{t-1}| - \mathbb{E}|z|) + \gamma z_{t-1}, \quad z_{t-1} = \frac{a_{t-1}}{\sigma_{t-1}}.$$

- **Change 1 – the log.** Modelling  $\log \sigma_t^2$  means  $\sigma_t^2 = e^{(\cdot)} > 0$  *automatically*: **no non-negativity constraints** on  $\omega, \alpha, \beta, \gamma$ . They can be any sign – a real practical relief.
- **Change 2 – the term  $\gamma z_{t-1}$ .** It carries the *sign* of the shock. The magnitude piece  $\alpha(|z| - \mathbb{E}|z|)$  is symmetric;  $\gamma z$  tilts it.
- If  $\gamma < 0$ : a negative  $z_{t-1}$  (bad news) *adds* to log-variance while a positive one subtracts – exactly the leverage effect. GARCH forces  $\gamma = 0$ .

# EGARCH – the fit and the news-impact curve



Fitted to daily S&P 500

$$\hat{\gamma} = -0.14 \quad (t = -19), \quad \hat{\beta} = 0.97.$$

- $\hat{\gamma} < 0$ , **overwhelmingly significant**: the leverage effect is real and strong.
- A one-s.d. *down* shock raises next-day volatility about  $1.5\times$  as much as an equal *up* shock.
- **Better fit, too**: EGARCH AIC = 17977 vs GARCH 18276 – the asymmetry earns its one extra parameter easily.

**The curve:** GARCH's response (blue) is a symmetric parabola in yesterday's shock; EGARCH's (red) is **tilted** – steeper on the loss side. Same  $|\text{shock}|$ , more volatility if it was a loss. (*Its simpler cousin, **GJR-GARCH**, adds a single  $\gamma a_{t-1}^2 \mathbf{1}_{\{a_{t-1} < 0\}}$  term to plain GARCH.*)



Putting it together: a comparison

# The four models at a glance

| Model         | Equation (vs GARCH)                                                                                              | What is new                       | Problem it fixes                                 |
|---------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------|
| GARCH(1, 1)   | $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$                                                  | baseline                          | clustering, long memory                          |
| IGARCH / EWMA | $\sigma_t^2 = (1 - \lambda)a_{t-1}^2 + \lambda\sigma_{t-1}^2$<br>( $\alpha + \beta = 1$ , $\omega = 0$ )         | persistence fixed at 1            | near-unit persistence; a simple, robust standard |
| GARCH-M       | $r_t = \mu + \delta \sigma_t + a_t$<br>(same $\sigma_t^2$ )                                                      | volatility enters the <i>mean</i> | risk-return premium is missing                   |
| EGARCH        | $\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2$<br>$+ \alpha( z_{t-1}  - \mathbb{E} z ) + \gamma z_{t-1}$ | log-variance + sign term          | leverage (asymmetry); no positivity constraints  |

## The same comparison, by behaviour and by fit

| Model         | Mean reverts?                  | Symmetric?           | Key parameter            | Fitted on daily S&P 500                           |
|---------------|--------------------------------|----------------------|--------------------------|---------------------------------------------------|
| GARCH(1, 1)   | yes ( $H \approx 37\text{d}$ ) | yes                  | $\alpha + \beta$         | $\hat{\alpha} = 0.12, \hat{\beta} = 0.86$         |
| IGARCH / EWMA | <b>no</b> ( $H = \infty$ )     | yes                  | $\lambda$                | $\hat{\lambda} \approx 0.93$ (RiskMetrics 0.94)   |
| GARCH-M       | yes                            | yes                  | $\delta$ (price of risk) | $\hat{\delta} = 0.11, t = 3.0$ (positive premium) |
| EGARCH        | yes                            | <b>no</b> (leverage) | $\gamma$ (asymmetry)     | $\hat{\gamma} = -0.14, t = -19$ ; AIC beats GARCH |

### One-line readings

**IGARCH** – “today’s vol is my forecast for every horizon.”   **GARCH-M** – “risk should be paid for.”   **EGARCH** – “losses frighten more than gains.”

## Which one should you reach for?

- **Start with GARCH(1,1).** It is the honest default and usually 80% of the answer.
- **Use IGARCH / EWMA** when you want a *simple, parameter-light* volatility tracker or a conservative risk monitor – and you do not need long-horizon mean reversion (RiskMetrics).
- **Use GARCH-M** when the *mean* is the object of interest and theory predicts a risk premium – asset-pricing and risk–return studies, often at weekly/monthly frequency.
- **Use EGARCH** (or GJR) when *asymmetry matters* – equity-index risk, option pricing, downside VaR. For equities this is almost always an improvement over plain GARCH.

### They are not mutually exclusive

The pieces compose: one can fit an **EGARCH-M with Student- $t$  innovations** – asymmetric variance, a risk premium in the mean, and fat tails – all at once. Start simple; add a feature only when a diagnostic or the economics demands it.

Beyond fixed parameters: regime switching

## A loose thread: why is persistence always so close to 1?

- Every GARCH we fit had  $\hat{\alpha} + \hat{\beta} \approx 0.98\text{--}0.99$ . Taken at face value, that says a volatility shock is *half*-remembered after two months.
- **But there is a competing explanation.** Suppose volatility is not one smoothly-decaying process, but **switches between distinct regimes** – long calm spells and shorter violent ones.

### Lamoureux & Lastrapes (1990)

If the true series has occasional **shifts in the variance level**, a single-regime GARCH – which assumes *one* constant long-run variance – mistakes those level-shifts for extreme persistence.

**Ignoring regimes inflates  $\hat{\alpha} + \hat{\beta}$  toward 1.**

- So far the *parameters* were fixed and only  $\sigma_t^2$  moved. Now we let the *model itself* change with an unobserved **state**. This is the doorway to **Tsay, Chapter 4: nonlinear models**.

# Two ways to switch regimes

## 1. Observed trigger – **threshold** models (Tong)

The regime is decided by a *variable you can see* crossing a threshold  $c$ . The **self-exciting TAR (SETAR)** uses the series' own lag:

$$r_t = \begin{cases} \phi_0^{(1)} + \phi_1^{(1)} r_{t-1} + a_t^{(1)}, & r_{t-d} \leq c \quad (\text{regime 1}), \\ \phi_0^{(2)} + \phi_1^{(2)} r_{t-1} + a_t^{(2)}, & r_{t-d} > c \quad (\text{regime 2}). \end{cases}$$

Different dynamics in each regime; the switch is *deterministic* given the data.

## 2. Hidden trigger – **Markov switching** (Hamilton, 1989)

The regime is an **unobserved state**  $S_t \in \{1, 2\}$  that follows a Markov chain. We never see it; we *infer* its probability from the data. This is our focus – it fits volatility regimes naturally.

Both live in Chapter 4: the conditional distribution is *nonlinear* – it depends on a state.

# The 2-state Markov-switching model

A hidden state drives which distribution generates the return

$$r_t \mid (S_t = s) \sim \mathcal{N}(\mu_s, \sigma_s^2), \quad s \in \{1, 2\},$$

and the state itself follows a **Markov chain** with transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \quad p_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \quad p_{i1} + p_{i2} = 1.$$

- **Persistence of a regime** is  $p_{ii}$ ; its **expected duration** is  $1/(1 - p_{ii})$  days.
- **Long-run share** of each regime is the stationary distribution  $\pi_1 = \frac{1-p_{22}}{2-p_{11}-p_{22}}$ ,  $\pi_2 = 1 - \pi_1$ .
- **Estimation:** we cannot see  $S_t$ , so we run the **Hamilton filter** to get  $\Pr(S_t = s \mid \text{data})$  and maximise the likelihood by **EM** – no new distributional magic, just the mixture done properly.



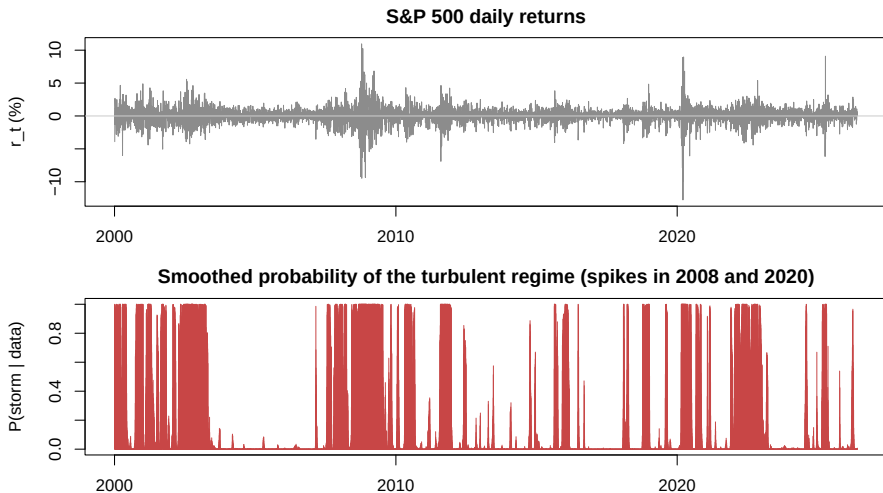
## S&P 500: two volatility regimes, estimated

### 2-state Markov-switching fit (daily S&P 500, 2000–2026)

| Regime       | Mean $\hat{\mu}_s$ | Vol $\hat{\sigma}_s$ (%/day) | Stay prob. $\hat{p}_{ss}$ | Expected duration |
|--------------|--------------------|------------------------------|---------------------------|-------------------|
| <b>Calm</b>  | +0.08              | 0.70                         | 0.99                      | $\approx 90$ days |
| <b>Storm</b> | −0.11              | 1.95                         | 0.97                      | $\approx 36$ days |

- **The regimes are economically distinct:** the calm state has *positive* drift and low volatility; the storm state has *negative* drift and  $2.8\times$  the volatility. Risk and return flip together.
- **Both are sticky** ( $p_{ss} \geq 0.97$ ): markets *stay* calm or panicked for weeks, not days.
- **Long-run split:**  $\approx 71\%$  calm,  $29\%$  storm. The mixture's unconditional s.d. is  $1.21\%$  – matching the sample  $1.22\%$ .

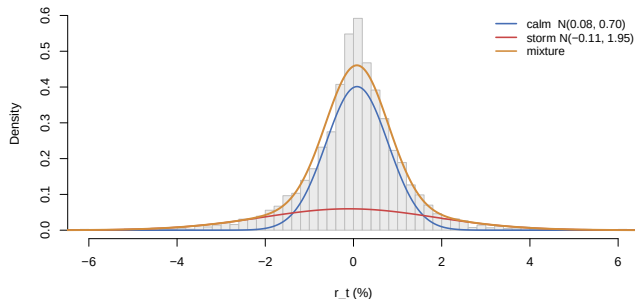
# Reading the market's hidden state through time



The smoothed probability  $\Pr(\text{storm} \mid \text{data})$  (bottom) is near zero for years, then **snaps to one** around 2008 and 2020 – the model dates the crises on its own, from returns alone.

# Fat tails, reinterpreted: a mixture of two Normals

One fat-tailed distribution = a mixture of two calm/storm Normals



- One **leptokurtic** return distribution = a calm Normal (narrow) plus a storm Normal (wide), blended by their time-shares.
- The peak comes from calm days; the **fat tails** come from storm days. Same fat tails we chased with Student-*t* – now with an *economic* story, not just a heavier density.

## Link back to GARCH

**MS-GARCH** lets each regime have its *own* GARCH. Once regimes absorb the level-shifts, the within-regime persistence falls – the Lamoureux–Lastrapes point, confirmed.

## Chapter 4 preview: nonlinear time series

## Where Chapter 4 goes

The common thread: the model depends on a *state* or a *threshold*

Linear ARMA/GARCH have one fixed set of dynamics. **Nonlinear** models let the dynamics themselves change – with an observed trigger, a hidden state, or a smooth blend.

| Model                              | Regime is chosen by                        | Switch        |
|------------------------------------|--------------------------------------------|---------------|
| <b>TAR / SETAR</b> (Tong)          | an observed threshold $r_{t-d} \lessgtr c$ | abrupt        |
| <b>STAR</b>                        | a smooth logistic function of $r_{t-d}$    | gradual       |
| <b>Markov switching</b> (Hamilton) | a hidden Markov state $S_t$                | probabilistic |
| <b>MS-GARCH</b>                    | hidden state $\times$ GARCH variance       | probabilistic |

- Chapter 4 also supplies **tests for nonlinearity** (e.g. Tsay's F-test, BDS) – decide *whether* you even need to leave the linear world before you do.
- **Takeaway:** volatility clustering (Ch 3) and regime switching (Ch 4) are two lenses on the same fact – financial risk is not constant, and not linear.

Practice

# Problem set

Five problems – attempt them before the next class

A separate handout, [GARCH\\_problemset.pdf](#), carries this session's exercises with *full worked solutions* at the end:

- ➊ **GARCH(1,1) at the edge** – half-life, the finite-kurtosis boundary, and the (surprising) number of days a 0.99-persistence forecast takes to revert.
- ➋ **IGARCH / RiskMetrics** – show EWMA is integrated and its forecast term structure is flat.
- ➌ **GARCH-M** – the price of risk, and why the daily estimate is economically fragile.
- ➍ **EGARCH** – quantify the leverage asymmetry from the news-impact function.
- ➎ **Markov switching** – regime durations, long-run shares, and the Lamoureux–Lastrapes persistence bias.

All numbers reproduce from `Day3_variants_code.R` and `Day3_regime_code.R`.

# Today in one slide

## The starting point

- GARCH(1,1) assumes: variance reverts, mean is risk-blind, shocks are symmetric.
- Each is an approximation the data strain against.

## IGARCH

- $\alpha + \beta = 1$ : shocks permanent, forecast flat. = EWMA / RiskMetrics ( $\lambda \approx 0.94$ ).

## GARCH-M

- $r_t = \mu + \delta\sigma_t + a_t$ : volatility priced into the mean. S&P  $\hat{\delta} = 0.11$  ( $t = 3$ ).

## EGARCH

- $\log \sigma_t^2$  with a  $\gamma z_{t-1}$  term: leverage, no positivity constraints. S&P  $\hat{\gamma} = -0.14$  ( $t = -19$ ), AIC beats GARCH.

## The empirical thread

On daily S&P 500 the leverage effect is the strongest signal ( $t = -19$ ) and buys the biggest fit gain; the risk premium is real but delicate; IGARCH is a convenience, not a discovery. **Match the model to the question.**



## Check your understanding

- 1 In IGARCH, why is there no unconditional variance, and what does that do to the  $h$ -step forecast?
- 2 RiskMetrics sets  $\lambda = 0.94$  and  $\omega = 0$ . Which GARCH quantities are being fixed, and to what?
- 3 In GARCH-M,  $\hat{\delta} > 0$ . Translate that into a sentence about expected returns on calm vs turbulent days.
- 4 In EGARCH, what does the term  $\alpha(|z_{t-1}| - \mathbb{E}|z|)$  do, and what does  $\gamma z_{t-1}$  add on top?
- 5 You fit GARCH and EGARCH; EGARCH's AIC is much lower and  $\hat{\gamma} < 0$ . What have you learned about the return series?

*(Answers:  $\alpha + \beta = 1$  so  $\omega/(1-\alpha-\beta)$  is infinite  $\Rightarrow$  the forecast never reverts (flat if  $\omega = 0$ );  $\beta = \lambda = 0.94$ ,  $\alpha = 1 - \lambda = 0.06$ , persistence = 1; expected return is higher when conditional volatility is higher – risk is paid for; the first is a symmetric magnitude response, the second tilts it by the shock's sign (leverage); volatility responds asymmetrically – bad news raises it more – so a symmetric GARCH is mis-specified.)*

## Next time

### Deeper into Chapter 4: nonlinear time series

Today's Markov switching was a first taste. Next we treat the **threshold (TAR/SETAR)** and **smooth-transition (STAR)** models in full, add formal **tests for nonlinearity**, and estimate a Markov-switching model on the S&P 500 start-to-finish. Then (Day 5) we go multivariate – **DCC** correlations across assets. *Volatility was never constant; it is not linear either.*

Questions?

## Appendix – reproduce everything in R

All figures and numbers come from `Day3_variants_code.R`; each variant is fitted by transparent hand-rolled MLE (`optim`) on the local `data/` folder.

```
r <- 100*diff(log(idx$SP500)); a <- r - mean(r)

# IGARCH / EWMA (RiskMetrics): sigma^2 = (1-lam)a^2 + lam*sigma^2
# GARCH-M:    r_t = mu + delta*sigma_t + a_t      (delta = price of risk)
# EGARCH:     log s2 = w + b*log s2_{-1}
#              + alpha(|z_{-1}|-E|z|) + gamma*z_{-1}
#              gamma < 0 ==> leverage (bad news raises vol more)
```

Fitted (daily S&P 500): IGARCH  $\hat{\lambda} = 0.93$ ; GARCH-M  $\hat{\delta} = 0.11$  ( $t = 3.0$ ); EGARCH  $\hat{\gamma} = -0.14$  ( $t = -19$ ), AIC 17977 vs GARCH 18276.